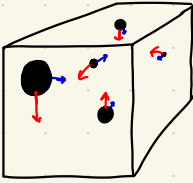


Goal: Plot an assortment of orbital bodies with randomly generated Velocity vectors in a bounded box. Go through each body and calculate the vector of gravitational force on it. Then add a visible red arrow for the velocity and a blue one for gravity.

e.g.



Functional Diagram

Equation 1.1

X # of bodies in y area
radius r of x bodies possible.
Let a be $0 < a \leq 1$.

What is the max amount r
can be to fit in a-y area?

$$a \cdot \frac{y}{2x} = \max(r)$$

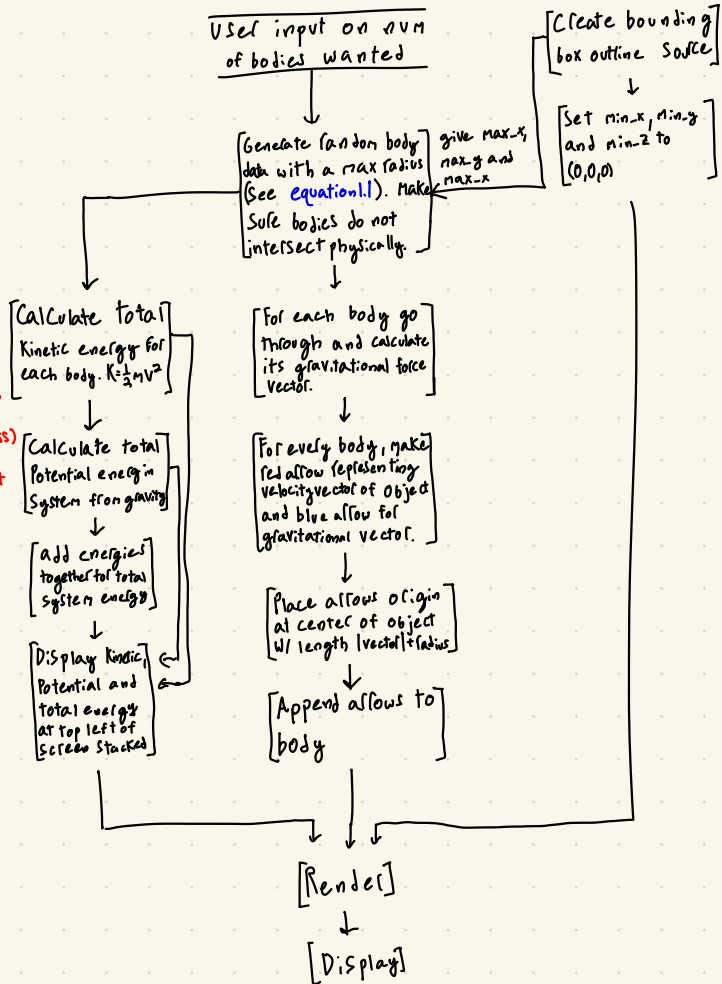
System Defaults

Unit of Mass = 1 solar mass (sunmass)

Unit of distance = 1 Astronomical unit

Unit of time = 1 day

Unit of energy = (Solar mass) $\cdot$ (AU)²



Custom Classes

Body
~Position Vector
~radius: Double
+distanceTo(Body): Double
+isIntersect(Body): Bool
+isInBounds(): Bool

NewtonianBody
-Velocity Vector
-Gravity Vector
-mass: Double
+GetKinetic(): Double
+GetPotential(bodies[]): Double

Object Diagram

[# of bodies]



NewtonianBody[]
list of randomly generated
NewtonianBody objects
w/ max possible radius
following equation 1.1

